Translation

SAT Math · Foundations to Mastery · Week of 18 October

Name: _____ Due: _____

How to use this sheet

This is four short sittings, not one long one. Four half-hours across the week will do more for you than one long Saturday, and you are far more likely to finish it.

- **Part A is no calculator.** None. It is 15 minutes and it keeps the last three weeks alive.
- **Part B asks you not to solve.** You write the equation and stop. The arithmetic is not the part costing you marks.
- When you get one wrong, write it in your error log: what you did, and what you should have done.

Sitting	Part	What it drills	Minutes
Sunday	A	Retention: signs, slope, systems	15
Tuesday	B	Setups only — do not solve	25
Thursday	C	The two traps	25
Saturday	D, E	Set up and solve; six mixed, timed	27

Part A — Retention

No calculator. 15 minutes.

SUNDAY

Three weeks of work sits behind these ten. Keep the brackets around every substitution.

1. $-9-(-4)=$ _____
2. Expand: $-3(x-5)$ _____
3. Expand and simplify: $-(4x-7)+x$ _____
4. A line passes through $(-2,7)$ and $(2,-1)$. What is its slope? _____
5. A line is given by $3x+4y=8$. What is its slope? _____
6. A line is given by $5x-2y=10$. What is the y-coordinate of its y-intercept? _____
7. Solve by substitution: $y=2x-3$ and $3x+y=12$. Give x and y.
   Answer: _____ *(write your answer)*
8. Solve by elimination: $4x+y=14$ and $2x-y=4$. Give x and y.
   Answer: _____ *(write your answer)*
9. Solve $2x+7<1$.
   Answer: _____ *(write your answer)*
10. Solve $5-3x\geq 20$.
    Answer: _____ *(write your answer)*

Part B — Setups only

Calculator allowed. 25 minutes.

TUESDAY

Do not solve any of these. For each one you write two things and then stop: first name the unknown on the *Let* line, then write the equation (or the pair of equations) on the line under it. No arithmetic. No answer.

You can do the arithmetic already. What costs you marks is building the equation, so this sitting practises only that. An equation you reached without naming the unknown does not count this week.

1. A moving company charges a flat \$75 for the van plus \$0.60 for every mile driven. A move cost \$153 in total. Write an equation for the number of miles.

   Let: _____

   Equation: _____

2. A printing shop charges an \$18 setup fee plus \$0.35 for each flyer. An order cost \$81. Write an equation for the number of flyers.

   Let: _____

   Equation: _____

3. A theatre sold 40 tickets and took \$395. Adult tickets cost \$12 and child tickets cost \$7. Write a *pair* of equations.

   Let: _____

   Equation: _____

4. A box holds 30 pens. Black pens cost \$1.20 each and blue pens cost \$0.80 each, and the whole box is worth \$30. Write a *pair* of equations.

   Let: _____

   Equation: _____

5. A train travels at a steady 56 miles per hour and covers 280 miles. Write an equation for the time taken.

   Let: _____

   Equation: _____

6. A tank holds 540 litres and a pump removes 18 litres each minute. At some moment 90 litres remain. Write an equation for the time the pump has run.

   Let: _____

   Equation: _____

7. In one class, 35% of the students chose Spanish, and 14 students chose Spanish. Write an equation for the number of students in the class.

   Let: _____

   Equation: _____

8. **Reminder: you are still not solving these.** A jacket is reduced by 20% and sells for \$68. Write an equation for the original price.

   Let: _____

   Equation: _____

9. A club raised its membership fee by 12%, and the new fee is \$504. Write an equation for the old fee.

   Let: _____

   Equation: _____

10. Three consecutive integers add to 84. Write an equation.

    Let: _____

    Equation: _____

11. Two consecutive odd integers add to 92. Write an equation.

    Let: _____

    Equation: _____

12. A rectangle has a length that is 5 cm more than twice its width, and its perimeter is 70 cm. Write an equation in the width alone.

    Let: _____

    Equation: _____

Part C — The two traps

Calculator allowed. 25 minutes.

THURSDAY

Two mistakes account for most of what goes wrong when you translate.

The reversal trap. "7 less than x" is $x-7$. Never $7-x$. The thing you start with is written first, and the amount taken away comes after the minus sign.

The direction trap. "Twice as many apples as pears", with a apples and p pears, is $a=2p$. The multiplier goes on the *smaller* side, because that is the side that needs building up to match.

1. Which expression represents "seven less than a number x"?

   A) $7-x$ B) $x-7$ C) $7x$ D) $\frac{7}{x}$

2. A number n decreased by 12 is equal to 5 times the number. Which equation says that?

   A) $12-n=5n$ B) $n-12=5n$ C) $n-12=n+5$ D) $12-n=\frac{n}{5}$

3. Which expression represents "9 subtracted from 4 times k"?

   A) $9-4k$ B) $4k-9$ C) $4(k-9)$ D) $4-9k$

4. A number a is 6 less than three times a number b. Which equation says that?

   A) $a=6-3b$ B) $a=3b-6$ C) $a=3(b-6)$ D) $6-a=3b$

5. A student is told "the cost c is \$8 less than the price p" and writes $c=8-p$. In one sentence, say why that is wrong, then write the correct equation.

   _____

   _____

6. A shelter has three times as many cats c as dogs d. Which equation says that?

   A) $c=3d$ B) $3c=d$ C) $c+3=d$ D) $c=d+3$

7. A theatre has 5 times as many standard seats s as premium seats p. Which equation says that?
   A) $5s = p$ B) $s = 5p$ C) $s = p + 5$ D) $p = s + 5$

8. A recipe uses twice as much flour f as sugar g, both measured in cups. Which equation says that?
   A) $f = 2g$ B) $2f = g$ C) $f = g + 2$ D) $fg = 2$

9. A class has 4 times as many juniors j as seniors n. A student writes $4j = n$. In one sentence, say why that is wrong, then write the correct equation.

   _____

   _____

10. A jar has half as many red marbles r as blue marbles b. A student writes $r = 2b$. In one sentence, say why that is wrong, then write the correct equation.

    _____

    _____

Part D — Set up and solve

Calculator allowed. 15 minutes.

SATURDAY

Now you do both halves. Name the unknown, write the equation, solve it, and then read your answer back against the story to check it makes sense.

1. A gym charges a one-time joining fee of $40 plus $29 for each month of membership. A member has paid $359 in total. For how many months has he been a member?

   Answer: _____ *(write your answer)*

2. A theatre sold 60 tickets and took $732. Adult tickets cost $15 and student tickets cost $9. How many student tickets were sold?

   Answer: _____ *(write your answer)*

3. Three consecutive integers add to 132. What is the greatest of the three?

   Answer: _____ *(write your answer)*

4. In a club, 45% of the members voted in an election, and 27 members voted. How many members does the club have?

   Answer: _____ *(write your answer)*

5. A rectangle has a length that is 3 cm less than twice its width, and its perimeter is 54 cm. What is its width, in cm?

   Answer: _____ *(write your answer)*

6. A tank holding 400 litres drains at 7 litres per minute. After how many minutes do 155 litres remain?

   Answer: _____ *(write your answer)*

Part E — Mixed, timed

Calculator allowed. 12 minutes, timed.

SATURDAY

Set a timer for 12 minutes. Answer every one — there is no penalty for a wrong answer, so a blank is a mark thrown away. Then check each answer by reading it back against the story.

1. A repair shop charges a flat \$25 inspection fee plus \$0.90 for each minute of labour. Which equation gives the total cost C, in dollars, of a repair taking t minutes?
   A) $C = 25t + 0.90$ B) $C = 0.90t + 25$ C) $C = 25.90t$ D) $C = 0.90(t + 25)$

2. A shop sells pens at \$2 each and markers at \$5 each. Tom bought 12 items and paid \$39. How many markers did he buy?
   Answer: _____ *(write your answer)*

3. A camp has 4 times as many campers c as counsellors n. Which equation describes this?
   A) $4c = n$ B) $c = 4n$ C) $c = n + 4$ D) $n = c + 4$

4. Twelve less than 5 times a number is 63. What is the number?
   Answer: _____ *(write your answer)*

5. A rectangle has width w and a length that is 4 more than three times the width. Which expression gives its perimeter?
   A) $8w + 8$ B) $4w + 4$ C) $6w + 4$ D) $3w + 4$

6. A jacket is marked down by 30% and sells for \$84. What was its original price, in dollars?
   Answer: _____ *(write your answer)*

Error log

Fill this in as you go.

Date	Q	What I did	What I should have done

Bring this sheet and your error log to the next session.